

TRAINING @ FORREST ONE

Session 1 of 12

Network Fundamentals + CompTIA Security+ SY0-701

Network Foundations: Networking Models & IP Addressing

How data moves across networks, from the signal on the wire to the subnet mask, and why every security control depends on it.

NETWORK

OSI/TCP-IP

SUBNETTING

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WELCOME · THE JOURNEY

Twelve sessions, two strands, one goal

Strand	Sessions	Focus
Network foundation	1–3	Models, addressing, switching, services
Security+ core	4–7	Concepts, threats, crypto, IAM
Security architecture	8–9	Network, endpoint, application, cloud
Operations & governance	10–12	Vuln mgmt, IR, GRC

Why networks first, and how today is built

Six chapters: the language of networks, the OSI model, physical and data link, network and transport, media and topologies, then IPv4 addressing and subnetting. Every firewall rule and IDS signature speaks this language, and every attack travels over it.



SESSION OBJECTIVES

By the end of this session you will be able to

- **Seven OSI layers** name each and its job
- **Encapsulation** a header at every layer
- **Physical layer** signals and media
- **MAC addressing** 48-bit hardware addresses
- **TCP against UDP** reliability or speed
- **Upper layers** session, presentation, app
- **Wired and wireless** choose suitable media
- **Topologies** star against mesh
- **An IPv4 mask** network against host
- **Subnets** classless prefixes

Ten objectives, one quiz

The session quiz at the end of these three hours tests every objective on this slide.



CHAPTER 1 · THE LANGUAGE OF NETWORKS

Why layers matter at all

- **Troubleshooting without structure** a user cannot reach a website, and you have no idea whether the fault is the cable, the switch, the route or the server
- **Seven layers, one job each** every fault becomes findable because you can test the layers one at a time
- **Eliminate as you go** test each layer in turn and rule it out
- **A map, not a product** nobody sells you a layer five; the value is in the thinking

The whole value of the model

Split one impossible problem into seven small ones, and troubleshooting stops being guesswork.



CHAPTER 2 · THE OSI MODEL

The seven layers

Layer	Name	Role
7	Application	The services users touch, such as web and email
6	Presentation	Encoding, compression and encryption
5	Session	Opens, manages and closes conversations
4	Transport	Delivers end to end, uses port numbers
3	Network	Logical addressing and routing
2	Data Link	Delivers frames on a local segment
1	Physical	Bits on the wire, light in fibre, radio waves



CHAPTER 2 · THE OSI MODEL

The journey of a message

Layer	What it adds	Address used
7 to 5 · Application to Session	The data itself	None
4 · Transport	Segmentation and ports	Port number
3 · Network	Logical addressing	IP address
2 · Data Link	Local delivery	Hardware address
1 · Physical	Signals	None

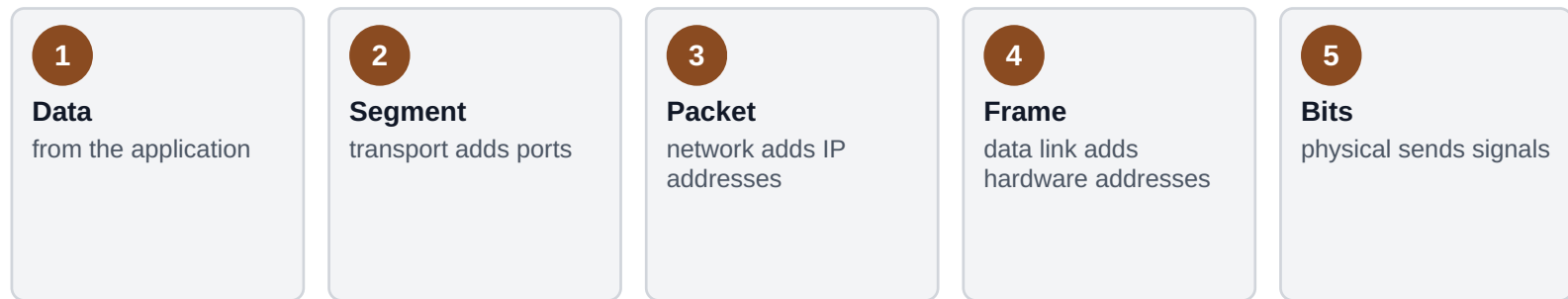
The receiver reverses the process

Each layer peels one header off on the way up, until the application has the data it asked for.



CHAPTER 2 · THE OSI MODEL

Encapsulation: same data, different name



Know the vocabulary precisely

TCP carries segments and UDP carries datagrams; both become packets at layer 3 and frames at layer 2.



CHAPTER 3 · PHYSICAL AND DATA LINK

The physical layer

- **Raw bitstream** carried between devices, with no understanding of meaning at all
- **Three signal types** electrical, light and radio
- **Copper** electrical signals, using voltage changes for ones and zeros
- **Fibre** light pulses, travelling further with far less interference
- **Wireless** radio waves through the air, which buys you mobility
- **This layer defines standards** cabling, connectors and signalling

First rule of troubleshooting

If layer one fails, nothing above it can possibly work. That is why the bottom of the stack is where you start.



CHAPTER 3 · PHYSICAL AND DATA LINK

Signals and their media

Signal	Medium	Notes
Electrical	Copper cable	Cheap and widely installed
Light	Fibre optic	Long distance, no interference
Radio	Air	Mobile, but shared airtime

Every medium is a trade

Cost, distance and interference. Those three factors will guide almost every cabling decision you make.



CHAPTER 3 · PHYSICAL AND DATA LINK

The data link layer

- **Node to node** delivers data between devices on the same local network
- **Media Access Control** 48 bits, written in hex, such as 00 1A 2B 3C 4D 5E
- **The frame** the unit of data at this layer
- **Two sublayers** Logical Link Control and Media Access Control
- **Error detection** a Cyclic Redundancy Check appended to the frame
- **Assigned, not trusted** set by the manufacturer, but changeable in software

Security note

Spoofing a hardware address is a software change, so layer 2 identity is never proof of identity.



CHAPTER 3 · PHYSICAL AND DATA LINK

The two data link sublayers

Sublayer	Responsibility	Faces
Logical Link Control	Identifies the layer 3 protocol, error checking	Up to Network
Media Access Control	Shares the medium, delivers by hardware address	Down to Physical

How the exam phrases it

Logical Link Control looks up towards the network layer; Media Access Control looks down towards the physical layer.



CHAPTER 3 · PHYSICAL AND DATA LINK

A dead link is almost always layer 1

- **The report** a user has lost connectivity completely
- **The temptation** dive into routing tables or firewall rules, because that feels clever
- **The discipline** check the cable, the port indicator and the link state first
- **Why start low** a dead link is usually physical, and physical faults are the most common and the cheapest to repair

Analyst's lens

Only once layer one is sound does it make sense to look higher up the stack.



CHAPTER 4 · NETWORK, TRANSPORT AND ABOVE

The network layer

- **Logical addressing** IP addresses, as opposed to hardware addresses
- **Between networks** it moves packets from one network to another
- **Routers** the devices that operate at this layer
- **Time to Live** drops by one at each router; at zero the packet is discarded
- **Multiple hops** data crosses the internet one router at a time
- **Adapts to failure** when a link fails, this layer chooses another path

The distinction that matters most

A hardware address identifies a device on a local segment. An IP address identifies a device across networks.



CHAPTER 4 · NETWORK, TRANSPORT AND ABOVE

The transport layer

- **Application to application** delivers data end to end between applications
- **Port numbers** so one machine runs web, mail and file sharing without confusion
- **Transmission Control Protocol** connection oriented, numbers every segment, retransmits losses
- **User Datagram Protocol** connectionless, no guarantee, no ordering, lower latency
- **The question to ask** does this application need reliability, or does it need speed?

Where each one belongs

Reliability for web, email and file transfer, where a missing byte matters. Speed for voice, video and lookups.



CHAPTER 4 · NETWORK, TRANSPORT AND ABOVE

TCP versus UDP

Feature	Transmission Control Protocol	User Datagram Protocol
Connection	Connection oriented	Connectionless
Reliability	Guaranteed, retransmits	Best effort
Ordering	Sequence numbered	Not guaranteed
Speed	Slower	Faster
Typical use	Web, email, file transfer	Voice, video, lookups

Why latency beats retransmission

If a packet of a video call arrives too late you do not want it retransmitted. A missing byte of a file transfer, you do.



CHAPTER 4 · NETWORK, TRANSPORT AND ABOVE

The upper layers: 5, 6 and 7

- **Session** establishes, manages and terminates conversations between applications
- **Session: dialogue control** whose turn it is to talk, and checkpoints in long transfers
- **Presentation** the translator, converting between data formats
- **Presentation: layer 6** handles compression and encryption
- **Application** where HTTP for the web and DNS for name resolution live

Exam trap

The application layer is the protocol layer, not the program itself. That distinction is examined directly.



DISCUSSION

**Name the layer first, then the control presents itself.
Why does that work?**

A web attack arrives at layer 7, so the answer is a web application firewall. A spoofed packet arrives at layer 3, so filter at the router. A rogue switch or a poisoned address table is layer 2, so use port security.

Each layer has its own attack and its own defence, and that is a direct consequence of each layer having one job.



BREAK
10 MINUTES

Back soon: the media your signals travel through, the topologies we arrange them in,
then IPv4 addressing and subnetting.



CHAPTER 5 · MEDIA AND TOPOLOGIES

Wireless transmission media

- **Radio waves** carry data through the air, with no cabling to install
- **Mobility** the obvious benefit of the wireless medium
- **Frequency bands** the band you choose is a real trade-off
- **Lower frequencies** travel further, pass through walls, carry less data
- **Higher frequencies** carry much more data over shorter distances
- **The limitation** wireless is shared, so every device competes for airtime

Design note

Performance drops as clients join the channel, and drops further with outside interference.



CHAPTER 5 · MEDIA AND TOPOLOGIES

Wired transmission media

- **Twisted pair** the cheapest and easiest to install, so it fills every office
- **Categories** rate capability, such as Category 5e and Category 6
- **Shielding** reduces interference from motors, lighting and other sources
- **Coaxial** higher frequencies than twisted pair; legacy networks and broadband
- **Fibre** carries light rather than electricity, so it is immune to interference
- **Fibre distance** spans the longest distances at the highest bandwidth

Rule of thumb

Running a link between two buildings? Fibre is almost always the answer.



CHAPTER 5 · MEDIA AND TOPOLOGIES

Choosing the right cable

Media	Signal	Best for
Twisted pair	Electrical	Office desktops, short runs
Shielded twisted pair	Electrical	Electrically noisy areas
Coaxial	Electrical	Legacy networks, broadband
Fibre optic	Light	Long distance, high bandwidth

Three questions choose the cable

How far? How much bandwidth? How much interference? Distance, bandwidth and interference decide.



CHAPTER 5 · MEDIA AND TOPOLOGIES

Connectors and transceivers

- **Transceiver** converts between signal types, so a copper port can talk to fibre
- **Registered Jack 45** the standard copper connector, on almost every desktop
- **Fibre connectors** Subscriber Connector and Lucent Connector
- **Small Form factor Pluggable** hot swappable, so media type changes without downtime
- **Media converters** bridge a copper segment to a fibre segment
- **A design decision** connector choice affects maintenance, not just speed

Why hot-swappable matters

You can change the media type on one port without taking the switch out of service.



CHAPTER 5 · MEDIA AND TOPOLOGIES

Star, mesh and hybrid

- **Star** every device connects to a central point, usually a switch
- **Star benefit** easy to troubleshoot, because each device has its own cable
- **Star weakness** the central point is a single point of failure
- **Full mesh** links every node to every other node, so any link can fail
- **Mesh trade** maximum resilience at maximum cost and complexity
- **Hybrid** meshes what needs resilience and stars the rest

One question, every topology

How much will you spend to survive a failure?



CHAPTER 5 · MEDIA AND TOPOLOGIES

Star against mesh

Topology	Strength	Weakness
Star	Simple: one fault, one device	Central point can fail
Full mesh	No single point of failure	Expensive and complex
Hybrid	Resilience where it matters	Needs careful design
Spine and leaf	Predictable latency	Requires more links

Spine and leaf, the data centre pattern

Every leaf switch connects to every spine switch, so any two devices sit the same number of hops apart.



CHAPTER 6 · IPV4 ADDRESSING AND SUBNETTING

IPv4 address fundamentals

- **32 bits** split into four groups of eight, and each group is an octet
- **Dotted decimal** 192.168.1.1, because the machine sees 32 binary digits
- **Each octet** runs from 0 to 255
- **The whole space** about 4.29 billion addresses
- **Why subnetting exists** nobody expected to run out of that space
- **Reserved ranges** withheld from public use, so private ranges are reusable

Start here every time

Thirty two bits, four octets, dotted decimal. Know the structure before you touch a mask.



CHAPTER 6 · IPV4 ADDRESSING AND SUBNETTING

Addresses to recognise on sight

Range	Purpose	Note
10.0.0.0/8	Private	Reusable inside any organisation
172.16.0.0/12	Private	Not routable on the public internet
192.168.0.0/16	Private	The common home and office range
127.0.0.1	Loopback	Always means the local machine
169.254.0.0/16	Automatic	Self-assigned when no server answers

A symptom, not a configuration

A 169.254 address means the machine asked for an address and nothing answered. Investigate the failure, not the address.



CHAPTER 6 · IPV4 ADDRESSING AND SUBNETTING

Subnet masks

- **A mask is 32 bits** a run of ones followed by a run of zeros
- **The ones** mark the network portion
- **The zeros** mark the host portion
- **255.255.255.0** means the first three octets are the network
- **Two addresses are reserved** the lowest is the network, the highest is the broadcast
- **Subnetting** divides one network into smaller segments

Count carefully here

Eight host bits give 256 values, minus the network and the broadcast: 254 usable hosts, in every subnet.



CHAPTER 6 · IPV4 ADDRESSING AND SUBNETTING

Worked example: reading a subnet mask

1

The address

192.168.1.50

2

The mask

255.255.255.0

3

The network

192.168.1.0

4

The host range

192.168.1.1 to .254

5

The broadcast

192.168.1.255

The pattern never changes

Find the network, count up from one, and stop one short of the broadcast. Do that quickly and you have the foundation.



Classless addressing

- **Classless Inter-Domain Routing** replaced the rigid class system
- **The slash** states how many bits are the network
- **/24** equals the mask 255.255.255.0
- **Counter-intuitive** a larger slash number means a smaller network
- **/30** gives just two addresses, which is why it links two routers
- **Variable Length Subnet Masking** lets each subnet differ in size, largest first

Why classless won

Classful addressing wasted whole blocks. CIDR lets the boundary fall at any bit, so the space you have is the space you use.



CHAPTER 6 · IPV4 ADDRESSING AND SUBNETTING

Prefix notation at a glance

Prefix	Subnet mask	Usable hosts
/24	255.255.255.0	254
/25	255.255.255.128	126
/26	255.255.255.192	62
/27	255.255.255.224	30
/30	255.255.255.252	2

The pattern to remember

Every extra prefix bit halves the addresses available, and two are always lost to the network and the broadcast.



QUIZ

Activity: quick subnet arithmetic

- What is the network address of 192.168.10.50/28?
- How many usable hosts in a /29 subnet?
- What is the CIDR notation for mask 255.255.255.240?
- Borrow 2 bits from a /24: how many subnets of what size?
- Which address can never be assigned to a host?

Block size shortcut: for /28 the block is 16, so 50 falls in the .48 to .63 range.



QUIZ

Check your understanding

- Open Quiz_Session_01.html in your browser.
- 20 questions, immediate feedback, and a topic tag on every question.
- Read each explanation: the distractors teach as much as the answer.
- Score 70% or better before moving on.

Every question maps to a session topic area, for example 1B IPv4 Addressing.



KEY TAKEAWAYS

- **Seven layers, one job each** which is what makes every fault findable by testing one layer at a time.
- **Encapsulation changes the name** data, segment, packet, frame, then bits on the medium.
- **Layer 2 is local** 48-bit hardware addresses deliver frames on one segment.
- **Layer 3 routes** IP addresses carry packets between networks, and Time to Live stops loops.
- **TCP guarantees, UDP is fast** choose reliability or latency for the application.
- **Media is a trade** distance, bandwidth and interference choose the cable.
- **Topology trades cost for resilience** decide where you can afford to fail.
- **The mask splits network from host** so a /24 gives 254 usable hosts, not 256.



REVISION CHECKLIST

Before next session

- **Retake the quiz** until you score 70% or better without guessing
- **Practise subnetting** ten /24 to /30 calculations by hand
- **Walk the stack** name each layer, its job and one protocol, out loud
- **Choose three cables** justify each with distance, bandwidth and interference
- **Read ahead** Session 2: switching, VLANs and spanning tree



NEXT SESSION

Session 2 — Switching, VLANs & Network Architectures

- MAC addresses and the switch forwarding table
- VLANs, trunking and 802.1Q tagging
- Spanning Tree Protocol and loop prevention

